

Schur Covariance Evaluation

A Principled Pseudo-Likelihood in High Dimensions

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Abstract

We introduce the **Schur pseudo-likelihood**, a one-parameter family $\ell_\gamma, \gamma \in [0, 1]$, that deforms the Gaussian likelihood by damping the cross-block coupling carried by its Schur complements. The endpoints are familiar — $\gamma=1$ is the exact joint likelihood, $\gamma=0$ is block composite (pseudo-)likelihood — and the interior is the new object: a continuous **coupling-strength** dial that the composite-likelihood, Vecchia, and Gaussian-Markov-random-field literatures do not have, because those methods regularize the *conditioning structure* (which variables to condition on) at full strength, an axis orthogonal to γ . The contribution is deliberately narrow and, we think, clean: not a new estimator but a *pseudo-likelihood with a single interpretable knob* that (i) recovers the power and conditioning the full likelihood loses in high dimensions, where it becomes — we show — worse than chance at ranking estimators; (ii) admits an exact characterization of what γ does, via a worked model we verify numerically: it recovers the true predictive law for every γ but maps to an inflated covariance that leaves the positive-definite cone below a coupling-dependent threshold — pinning down precisely why ℓ_γ is a device for *scoring* or *transforming* a covariance rather than for estimating one by free maximization; and (iii) is **broadly applicable** — wherever a Gaussian block likelihood is used (spatial statistics, graphical models, longitudinal data, any composite-likelihood setting), and even, via the identical algebra, in portfolio allocation, where the same γ interpolates hierarchical risk parity and minimum-variance optimization. We show in simulation that an interior γ strictly dominates both endpoints when the coupling is partially reliable, that a geodesic (affine-invariant) damping is stable where the linear one is not, and that the high-dimensional inversion catastrophe the construction avoids is borne out on real equity returns.

1. The high-dimensional failure of the likelihood

For a zero-mean Gaussian the per-sample log-likelihood is $\ell(\Sigma; x) = -\frac{1}{2}(\log \det \Sigma + x^\top \Sigma^{-1}x)$, equivalently the **covariance-likelihood** form in the sample covariance S , $\ell(\Sigma; S) = -\frac{1}{2}(\log \det \Sigma + \text{tr}(\Sigma^{-1}S))$ — proportional to the Wishart log-likelihood of S given Σ and, up to constants, the negative Stein/entropy loss. Both terms are governed by the *smallest* eigenvalues of Σ : $\log \det \Sigma = \sum \log \lambda_i \rightarrow -\infty$ and Σ^{-1} blows up as $\lambda_{\min} \rightarrow 0$. When $p/n \rightarrow c > 0$ (the Marčenko–Pastur / spiked regime) those eigenvalues collapse toward zero and are unidentifiable, so the likelihood stakes its verdict on noise. Empirically (research/metric_power.py) the held-out likelihood drops to ≈ 0.45 — *below chance* — at correctly ordering two estimates of *known* relative quality, while inversion-free judges hold ≈ 0.85 . This is the evaluation-side image of Markowitz “error maximization” in $w \propto \Sigma^{-1}1$: the same Σ^{-1} , the same fragility. The remedy is to stop trusting the raw inverse — the question is how, with one interpretable knob.

2. The exact Schur factorization of the Gaussian likelihood

Partition the p variables into K ordered blocks $x = (x_1, \dots, x_K)$. The joint density factorizes *exactly* into block conditionals, $p(x) = \prod_k p(x_k | x_{\setminus \{k\}})$, and for a Gaussian each is itself Gaussian with

- conditional covariance = **Schur complement** $S_{\setminus k} = \Sigma_{\setminus k \setminus k} - \Sigma_{\setminus k, k} \Sigma_{k, k}^{-1} \Sigma_{k, \setminus k}$,
- conditional mean $\mu_{\setminus k | k} = \mu_{\setminus k} + \Sigma_{\setminus k, k} \Sigma_{k, k}^{-1} (x_k - \mu_k)$.

Hence $\log \det \Sigma = \sum_k \log \det S_{\setminus k}$ and the quadratic form splits blockwise — the block-LDL^T / block-Cholesky factorization, the engine of Vecchia, GMRF, and nested-dissection elimination. **No approximation has been made yet:** only a re-expression of the same likelihood as a sum of block-conditional terms, each inverting only the conditioning block $\Sigma_{\setminus \{k\}, \setminus \{k\}}$.

3. The Schur pseudo-likelihood

We introduce a single **coupling-strength** parameter $\gamma \in [0, 1]$ that damps the Schur complement and the conditional regression in lockstep:

$$\begin{aligned} S_{\mathbf{k}}(\gamma) &= \Sigma_{\mathbf{k}\mathbf{k}} - \gamma \cdot \Sigma_{\{\mathbf{k}, <\mathbf{k}\}} \Sigma_{\{<\mathbf{k}, <\mathbf{k}\}}^{-1} \Sigma_{\{<\mathbf{k}, \mathbf{k}\}} = (1-\gamma) \Sigma_{\mathbf{k}\mathbf{k}} + \gamma S_{\mathbf{k}} \\ \mu_{\{\mathbf{k}|<\mathbf{k}\}}(\gamma) &= \mu_{\mathbf{k}} + \gamma \cdot \Sigma_{\{\mathbf{k}, <\mathbf{k}\}} \Sigma_{\{<\mathbf{k}, <\mathbf{k}\}}^{-1} (\mathbf{x}_{\{<\mathbf{k}\}} - \mu_{\{<\mathbf{k}\}}) \end{aligned}$$

$$\ell_{\gamma}(\Sigma; \mathbf{x}) = \Sigma_{\mathbf{k}} \log N(\mathbf{x}_{\mathbf{k}}; \mu_{\{\mathbf{k}|<\mathbf{k}\}}(\gamma), S_{\mathbf{k}}(\gamma)).$$

γ	ℓ_{γ} is	character
1	the exact joint Gaussian likelihood	most powerful, fragile (inverts all of Σ)
$(0, 1)$	a tunable bridge that <i>changes the block covariances</i>	inverts only conditioning blocks; better-conditioned
0	the block composite likelihood	robust; product of block marginals

What kind of object is ℓ_{γ} ? At $\gamma=1$ it is the joint log-density; at $\gamma=0$ it is a product of block marginals — a composite likelihood in the sense of Besag/Lindsay/Varin–Reid–Firth: a sum of proper component log-densities that does not itself integrate to one. For $\gamma \in (0, 1)$ each term $\log N(\mathbf{x}_{\mathbf{k}}; \mu(\gamma), S_{\mathbf{k}}(\gamma))$ is still a *genuine* Gaussian log-density (a strictly proper local score whenever $S_{\mathbf{k}}(\gamma) \succ 0$), but the damped conditionals are mutually inconsistent — they are not the conditionals of any single joint Gaussian — so ℓ_{γ} is a **pseudo-likelihood**, not a normalized joint. This is the composite-likelihood bargain: trade the joint-density property for tractable, well-conditioned components, and γ controls how much of the cross-block information the marginals discard is dialed back in.

Three uses, only two of them well-behaved. (a) *Score* a fixed estimate Σ by $\ell_{\gamma}(\Sigma; S_{\text{test}})$. (b) *Transform* a given Σ by replacing each conditional covariance with $S_{\mathbf{k}}(\gamma) = (1-\gamma)\Sigma_{\mathbf{k}\mathbf{k}} + \gamma \hat{S}_{\mathbf{k}}$ — a structured (block-conditional) **shrinkage** of the coupling toward block-diagonal with intensity $1-\gamma$, the basis-dependent counterpart of spectral shrinkage (§6); always PSD for $\gamma \in [0, 1]$ and the operation `SchurCovariance` and `Schur allocation` perform. (c) *Estimate* Σ by maximizing $\ell_{\gamma}(\Sigma; S)$ — which §4 shows is **degenerate** for $\gamma < 1$. The clean reading is that ℓ_{γ} is a scoring/transform device, not a free estimation objective.

4. What γ does, exactly: a worked model

To see precisely what damping does — and why estimation by free maximization fails while scoring and transformation do not — take the smallest non-trivial case and solve it in closed form. Let $\mathbf{x}_1 \sim N(0, \mathbf{a}), \mathbf{x}_2 = \mathbf{b}\star \mathbf{x}_1 + \epsilon, \epsilon \sim N(0, \mathbf{s}\star)$, two scalar blocks, so the true conditional law of $\mathbf{x}_2$ given $\mathbf{x}_1$ has regression $\mathbf{b}\star$ and Schur complement (conditional variance) $\mathbf{s}\star$. For a candidate $\Sigma = (\mathbf{A}, \mathbf{C}, \mathbf{D})$ the damped conditional has effective coefficient $\beta = \gamma \mathbf{C}/\mathbf{A}$ and variance $S(\gamma) = \mathbf{D} - \gamma \mathbf{C}^2/\mathbf{A}$. Maximizing the population (expected) ℓ_{γ} over $(\mathbf{A}, \mathbf{C}, \mathbf{D})$ gives (verified numerically in `research/schur_likelihood_theory.py`):

Proposition (two-block Gaussian). For every $\gamma \in (0, 1]$ the maximizer recovers the *true predictive law* — $\beta = \mathbf{b}\star$ and $S(\gamma) = \mathbf{s}\star$ — but the *implied covariance* is **inflated**: $\hat{\mathbf{A}} = \mathbf{a}, \hat{\mathbf{C}} = \Sigma_{12}\star / \gamma, \hat{\mathbf{D}} = \Sigma_{22}\star + \mathbf{b}\star^2 \mathbf{a} (1/\gamma - 1)$. This Σ is positive-definite iff $\gamma^2 (1 - \rho^2) > (1 - \gamma) \rho^2$, where $\rho^2 = \mathbf{b}\star^2 \mathbf{a} / (\mathbf{b}\star^2 \mathbf{a} + \mathbf{s}\star)$ is the block coupling (the conditional R^2). Below the threshold $\gamma_{\min}(\rho^2)$ the free maximizer leaves the PSD cone.

The numerics ($\mathbf{a}=1, \mathbf{b}\star=0.8, \mathbf{s}\star=0.5$, so $\rho^2=0.561$) show $\beta=0.800$ and $S(\gamma)=0.500$ for every γ while $\hat{\mathbf{C}}$ runs $0.80 \rightarrow 8.0$ as $\gamma:1 \rightarrow 0.1$, and the maximizer is PSD only above $\gamma_{\min} = 0.660$.

The general case (verified in `research/schur_likelihood_theory.py` by an exact closed-form expected ℓ_{γ} and independent local search):

Proposition (general blocks). For an arbitrary ordered partition the population ℓ_{γ} maximizer recovers *every block's true conditional law* — effective regression $G_{\mathbf{k}}(\gamma) = \mathbf{B}_{\mathbf{k}}\star$, damped conditional covariance $S_{\mathbf{k}}(\gamma) = S_{\mathbf{k}}\star$ — for every $\gamma \in (0, 1]$, and the implied joint covariance is inflated

sequentially: block k 's cross-block coupling scales by $1/\gamma$ through the (already inflated) conditioning covariance $\Sigma_{\{<k, <k\}}$. For **two vector blocks** the PSD condition is exactly $\gamma^2 (1 - \rho^2_{\max}) > (1 - \gamma) \rho^2_{\max}$, where $\rho^2_{\max}$ **is the largest squared canonical correlation between the blocks** — the scalar ρ^2 generalized to the dominant coupling mode. For K blocks the inflation, and hence the PSD threshold, **compound along the chain**.

The verification confirms each clause: the recovered $(G_k(\gamma), S_k(\gamma))$ equal the truth at every γ ; the local search finds nothing above the closed form (max-gap 0); and for a random vector two-block problem with $\rho^2_{\max}=0.115$ the predicted threshold $\gamma_{\min}=0.302$ matches the exact PSD boundary (SPD at $\gamma=0.32$, not at $\gamma=0.28$). Three consequences, now established beyond the scalar case:

1. **Damping is undone by rescaling.** ℓ_γ “knows” the true conditional distribution at any γ ; it simply expresses it through a coefficient scaled by $1/\gamma$. So *fitting* Σ to maximize ℓ_γ overcompensates — the implied joint inflates and, for strong enough coupling relative to γ , ceases to be a covariance at all. ℓ_γ **is degenerate as a free estimation objective for $\gamma < 1$** ; this is exactly why the same uses are to *score* a covariance or to *transform* one (shrink the coupling, §3b) — operations that hold Σ fixed rather than letting it absorb the damping.
2. ℓ_γ **is a bias–variance dial on the scoring rule.** Because the expected score is maximized at the *inflated* $\Sigma(\gamma)$, not at $\Sigma^\star$, ℓ_γ is for $\gamma < 1$ **not strictly proper** for Σ : it carries a (mild, for γ near 1) bias toward inflation. Its advantage is variance: the full likelihood ($\gamma=1$) is properly centered but, through Σ^{-1} and $\log \det$, has exploding variance in high dimensions; damping trades a little properness for a large reduction in score variance. The empirically optimal evaluation γ (§5) is the one whose bias²+variance for *ranking* is smallest.
3. **The usable range shrinks with coupling.** $\gamma_{\min}(\rho^2)$ increases from 0 (weak coupling, any γ safe) to 1 (strong coupling, only the full likelihood stays PSD). This is the same near-singular Schur-complement regime in which linear damping destabilizes and geodesic damping does not (§9) — two faces of one fact: strong coupling is where naïve interpolation of the conditioning is most dangerous.

5. The discrimination efficiency, exactly, and the Godambe boundary

§4(2) called γ a bias–variance dial on the scoring rule. This makes that statement exact, and locates — honestly — where the high-dimensional advantage actually comes from.

The γ -Schur operator. The score is a quadratic form in the data, $\ell_\gamma(\Sigma; x) = -\frac{1}{2} [Ld_\gamma(\Sigma) + x^T W_\gamma(\Sigma) x] + \text{const}$, where $W_\gamma(\Sigma) = \Sigma_k T_k^T S_k(\gamma)^{-1} T_k$ assembles the damped block conditionals ($T_k = [-\gamma \hat{G}_k, I, 0]$ picks block k minus its γ -damped regression on earlier blocks) and $Ld_\gamma = \Sigma_k \log \det S_k(\gamma)$. At $\gamma=1$, $W_1(\Sigma) = \Sigma^{-1}$ *exactly* (the block-LDL^T reconstruction of the precision; verified to $1e-15$). So the family deforms the precision continuously from Σ^{-1} ($\gamma=1$) to $\text{block-diag}(\Sigma_{kk}^{-1})$ ($\gamma=0$).

Discrimination SNR (exact, Gaussian). For two estimates A, B and truth Σ_t , the per-test-sample score gap $d(x) = \ell_\gamma(A; x) - \ell_\gamma(B; x)$ under $x \sim N(0, \Sigma_t)$ has, by the Gaussian quadratic-form moments, $\mu_\gamma = -\frac{1}{2} [\Delta Ld + \text{tr}(\Delta W \Sigma_t)]$ and $\sigma^2_\gamma = \frac{1}{2} \text{tr}((\Delta W \Sigma_t)^2)$, with $\Delta W = W_\gamma(A) - W_\gamma(B)$. Hence n_{test} samples rank A above B with probability $\approx \Phi(\sqrt{n_{\text{test}}} \cdot \mu_\gamma / \sigma_\gamma)$: the per-sample SNR $\mu_\gamma / \sigma_\gamma$ is the discrimination efficiency (closed form matches Monte Carlo to three figures).

The honest twist — under test noise alone, $\gamma=1$ is optimal. This SNR holds the estimates fixed and averages over test draws. In that model $\gamma=1$ maximizes the SNR in every regime we checked (well- and ill-conditioned candidates alike) — Neyman–Pearson — and damping only adds the §4 bias. **So the high-dimensional interior optimum is *not* a test-sample-noise effect.**

Where it does come from — estimate instability $\times$ shared unidentifiable structure. The score depends on the estimate through $W_\gamma(\Sigma)$. At $\gamma=1$, $W_1 = \Sigma^{-1}$ has sensitivity $\partial \Sigma^{-1} \sim \Sigma^{-1} \otimes \Sigma^{-1} \sim 1/\lambda_{\min}^2$: when the estimates carry a shared, unidentifiable, ill-conditioned component (e.g. spurious noisy cross-block coupling), a fresh data draw moves W_1 wildly and the score's verdict flips — power collapses to chance. Damping *bounds* that sensitivity (W_γ inverts only conditioning blocks). In the regime where the discriminating signal lives in the within-block correlations and the cross-block coupling is shared noise (`research/metric_power.py`, `research/schur_godambe.py`), an interior γ strictly beats both endpoints — power **0.84** (crude cross-damping)

/ **0.71** (the principled w_γ) at the optimum versus ≈ 0.45 at $\gamma \in \{0, 1\}$. The bias–variance dial of §4(2) is thus over the *estimate*, not the test sample.

The Godambe boundary. Whether ℓ_γ is even a valid estimating equation settles its status. The γ -Schur score $U_\gamma = \nabla_{\Sigma} \ell_\gamma$ is unbiased at the truth — $E_t[U_\gamma(\Sigma_t)] = 0$ — *only at the endpoints*: $\gamma=1$ (the full Fisher score) and $\gamma=0$ (the correctly specified block marginals, for their identified parameters). In the interior the expected score gradient is nonzero (verified: $\|\nabla E \ell_\gamma(\Sigma_t)\| = 0$ at $\gamma \in \{0, 1\}$ and grows through the interior), because the damped conditionals are misspecified — the inflation of §4. Standard (unbiased-estimating-function) Godambe theory thus applies only at the endpoints: $\gamma=0$ carries the classical composite-likelihood efficiency loss $G_0 = H_0 J_0^{-1} H_0 \leq I_{\text{full}}$ (Varin–Reid–Firth 2011); $\gamma=1$ is Fisher-efficient. The interior is neither an efficient estimator nor an unbiased score — it is a *tempering*, justified by the discrimination SNR above. This is the rigorous form of “ ℓ_γ is a scoring/transform device, not a free estimation objective.”

6. Schur damping is a structured shrinkage (and is complementary to spectral shrinkage)

The transform use (§3b) shrinks each conditional covariance from its full Schur complement $\hat{\Sigma}_k$ toward the unconditional block Σ_{kk} with intensity $1-\gamma$. Since $\hat{\Sigma}_k \leq \Sigma_{kk}$, this *raises* the conditional covariance, lifting the small eigenvalues of every block-sized inverse ℓ_γ performs — a James–Stein-style shrinkage applied *blockwise to the conditioning*. It is therefore **orthogonal and complementary to spectral shrinkage**: Ledoit–Wolf and the rotationally-invariant estimators (Bun, Bouchaud & Potters 2017) regularize the *eigenvalues* (basis-free); Schur damping regularizes the *cross-block coupling* (basis-dependent). Both improve conditioning, along different geometry, and can be composed: γ is to the block-conditional factorization what shrinkage intensity is to the spectrum.

7. The unifying role: one knob across many constructions

The same ℓ_γ places a scattered set of objects on **two axes of the same block factorization**:

- **Axis A — conditioning structure (sparsity)**: for each block, *which / how many* earlier blocks it conditions on. Full (dense) $\leftrightarrow$ none (block-diagonal). The axis Vecchia, GMRF, banding/tapering and nested dissection regularize along.
- **Axis B — coupling strength γ** : *how strongly* the chosen conditioning is trusted (§3). Full ($\gamma=1$) $\leftrightarrow$ none ($\gamma=0$). The axis introduced here.

	$\gamma = 1$ (full trust)	$\gamma \in (0,1)$	$\gamma = 0$ (no trust)
full conditioning (dense)	exact Gaussian likelihood = MVO $w \propto \Sigma^{-1}1$	SCHUR PSEUDO-LIK (this work) Schur + sparsity	block composite likelihood = HRP-style allocation
sparse conditioning (Vecchia/GMRF)	block-Vecchia / GMRF likelihood (exact conds.)		(block-marginal, same column)
nested dissection	= Schur-complement elimination of the (sparse) blocks		

- **Vecchia / GMRF / nested dissection** (Vecchia 1988; Katzfuss–Guinness 2021; Pan et al. 2024; Rue–Held 2005; George 1973) use the *exact* conditioning ($\gamma=1$) and regularize by **sparsity**. The Schur pseudo-likelihood adds the orthogonal **strength** axis; the two compose.
- **Composite / pseudo-likelihood** (Besag 1975; Lindsay 1988; Varin–Reid–Firth 2011) is the $\gamma=0$ column; ℓ_γ is its one-parameter deformation back toward the joint likelihood.
- **Portfolio allocation, the same algebra transported**: minimum-variance ($w \propto \Sigma^{-1}1$) is the $\gamma=1$ full-conditioning corner, hierarchical risk parity (López de Prado 2015/2016) the $\gamma=0$ corner, and **Schur complementary allocation** (Cotton 2024) damps *the same Schur complement with the same γ* between them. The Schur pseudo-likelihood is its evaluation/estimation mirror — identity, not analogy; Antonov–Lipton–de Prado (2024) on HRP stability explains the interior dominance.

One sentence: γ is one knob that, at its endpoints, recovers the joint likelihood, composite likelihood, block-Vecchia/GMRF likelihoods, MVO and HRP — and in its interior is the common bridge all of these lack.

8. Broad applicability

The construction needs only three ingredients — a Gaussian (or Gaussian-pseudo) likelihood, a block partition, and the conditional/Schur factorization of §2 — so the coupling-strength knob is **generic to Gaussian-block models**, not specific to covariance estimation or finance:

- **Spatial statistics / Gaussian processes.** Vecchia and GMRF likelihoods already condition each location on a neighborhood; γ adds a robustness dial on *how strongly* those neighbor relations are trusted — orthogonal to, and composable with, the existing choice of neighborhood size/ordering.
- **Gaussian graphical models / precision estimation.** With blocks as cliques or neighborhoods, γ damps the conditional-dependence strength, a continuous relaxation between the full penalized likelihood and node-/block-wise pseudo-likelihood (Besag; Meinshausen–Bühlmann).
- **Longitudinal / panel / state-space models.** With blocks as time slices, γ interpolates between treating slices independently ($\gamma=0$) and the full temporal joint ($\gamma=1$) — a tempering of the temporal coupling.
- **General composite likelihood.** Anywhere composite likelihood is used for tractability (Varin–Reid–Firth’s survey spans genetics, networks, multivariate survival, image analysis), γ is a generic dial from the composite ($\gamma=0$) toward the full ($\gamma=1$) likelihood, with §4’s caveat that it is to be used for scoring/tempering, not free estimation.
- **High-dimensional model scoring.** As a *score*, ℓ_γ is a high-dimension-robust, nearly-proper alternative to the log score for any of the above — the use we test directly here.

What we prove is the two-block characterization (§4) and demonstrate the covariance-evaluation case (§9); the wider applicability is a claim about the *mechanism* (damping a conditional factorization shared by all these models), flagged as such, not a measured result in each domain.

9. Geometry, and results

Linear vs geodesic damping. Damping interpolates each conditional covariance between Σ_{kk} and S_k . Linear damping is $S_k(\gamma) = (1-\gamma)\Sigma_{kk} + \gamma S_k$; **geodesic** damping moves along the affine-invariant SPD geodesic $\Sigma_{BD} \#_\gamma \Sigma = \Sigma_{BD}^{\{1/2\}} (\Sigma_{BD}^{\{-1/2\}} \Sigma \Sigma_{BD}^{\{-1/2\}})^{\gamma} \Sigma_{BD}^{\{1/2\}}$ (the Fisher–Rao geodesic, reusing the `GeodesicEwaCovariance` step), lifting small eigenvalues *multiplicatively*. They agree to first order and at the endpoints but differ near a singular Schur complement.

Results (reproducible; rankings are ensemble-dependent — see `papers/evaluation_and_generation_review.md`):

- γ **traces a regime-dependent optimum** (`research/metric_power.py`). Power climbs to $\gamma=1$ when the coupling is reliable (Neyman–Pearson); falls toward $\gamma=0$ in the high-dimensional noisy regime; and at *partial* reliability an **interior γ strictly dominates both endpoints** (≈ 0.84 vs ≈ 0.43) — the bias–variance optimum of §4(2), and the likelihood analogue of Schur allocation beating both HRP and MVO.
- **Geodesic stability.** Near a singular Schur complement (strong coupling — the $\gamma < \gamma_{\min}$ danger zone of §4(3)) linear damping dips (≈ 0.64) while geodesic damping holds full power.
- **The motivating failure, economically** (Ken French 100 portfolios, rolling minimum-variance OOS volatility, $p=100$, $n=60$; `research/oos_equity.py`). Not a test of ℓ_γ as an equity estimator — we are explicit about that — but a confirmation of the catastrophe it avoids: inverting/unregularized estimators blow up (annualized vol 9.6 empirical, 6.8 Huber, 1.3×10^6 Tyler) while well-conditioned shrinkage stays at 0.141 (vs 0.129 oracle, 0.228 equal-weight). In high dimensions the reward is for *conditioning* — for not trusting the raw inverse, which is what $\gamma < 1$ does.

10. Limitations and open problems

- **Estimate-variance theorem.** §5 gives the exact test-sample discrimination SNR (where $\gamma=1$ is optimal) and the Godambe boundary (the score is unbiased only at $\gamma \in \{0, 1\}$); the endpoint efficiency loss is classical. What is still empirical is the *estimate-instability* SNR — the variance of the score over training draws, driven by $\partial W_\gamma / \partial \Sigma \sim 1/\lambda_{\min}^2$ at $\gamma=1$ — whose closed form under a spiked/Marčenko–Pastur model would turn the interior optimum (now demonstrated) into a theorem and predict γ^* .

- **Compounded threshold.** §4 has the two-block $\rho^2_{\max}$ PSD form and numerical evidence that the K -block threshold compounds along the chain, but not yet a closed-form chain formula.
- **Optimal γ .** The evaluation/tempering optimum $\gamma^*(p, n, \text{spectrum}, \text{blocks})$ is characterized only empirically; a plug-in rule from the effective rank / coupling ρ^2 is open. (We are wary of a *trained* selector after the sibling recommender failed to generalize across novel generative families — `research/oos.py`.)
- **Block ordering and choice,** and composition with the Axis-A sparsity choice (the “Schur + sparsity” cell), remain to be studied; like Vecchia, ℓ_γ depends on the partition/ordering.
- **Scaling.** Recursive block inversion gives $O(p \cdot b^2)$; a Russell-scale ($p \approx 2000$) real-data study, where the separation should be largest, is set up to run.
- **Falsification.** The thesis predicts the interior- γ dominance should *vanish* when the coupling is fully reliable (LKJ, large η , low dimension $\rightarrow \gamma^* \rightarrow 1$) or pure noise (hard Marčenko–Pastur bulk $\rightarrow \gamma^* \rightarrow 0$); an interior optimum in a known-fully-reliable regime would contradict it.

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References

López de Prado (2015/2016); Antonov, Lipton & López de Prado (2024, SSRN 4748151); Cotton (2024, arXiv:2411.05807); Besag (1975); Lindsay (1988); Varin, Reid & Firth (2011); Vecchia (1988); Katzfuss & Guinness (2021); Pan, Abdulah, Genton & Sun et al. (2024); Rue & Held (2005); George (1973); Meinshausen & Bühlmann (2006); Ledoit & Wolf (2004, 2012); Bun, Bouchaud & Potters (2017); James & Stein (1961); Gneiting & Raftery (2007); Scheuerer & Hamill (2015). Full annotated lists in `papers/evaluation_and_generation_review.md` and `papers/covariance_evaluation.md`.